CO3 AT2

Level 2 Exam

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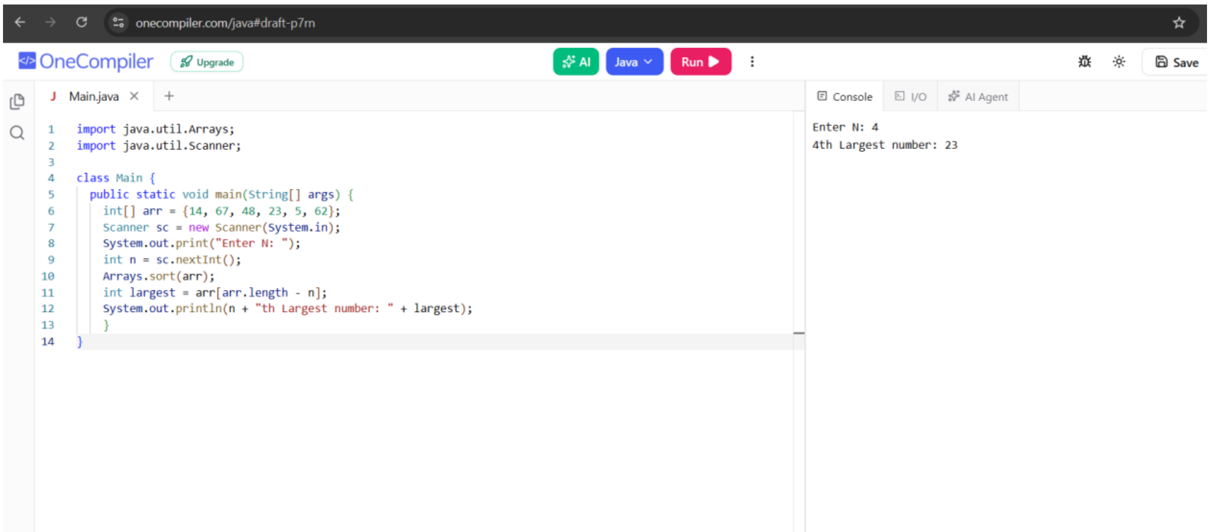
COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

1. Write a Program to Find the Nth Largest Number in a array. Sample Input: List : {14, 67, 48, 23, 5, 62} N = 4 Sample Output: 4 th Largest number: 23

```
import java.util.Arrays;

import java.util.Scanner;


class Main {

  public static void main(String[] args) {


    int[] arr = {14, 67, 48, 23, 5, 62};


    Scanner sc = new Scanner(System.in);

    System.out.print("Enter N: ");

    int n = sc.nextInt();


    Arrays.sort(arr);


    int largest = arr[arr.length - n];


    System.out.println(n + "th Largest number: " + largest);

  }
```

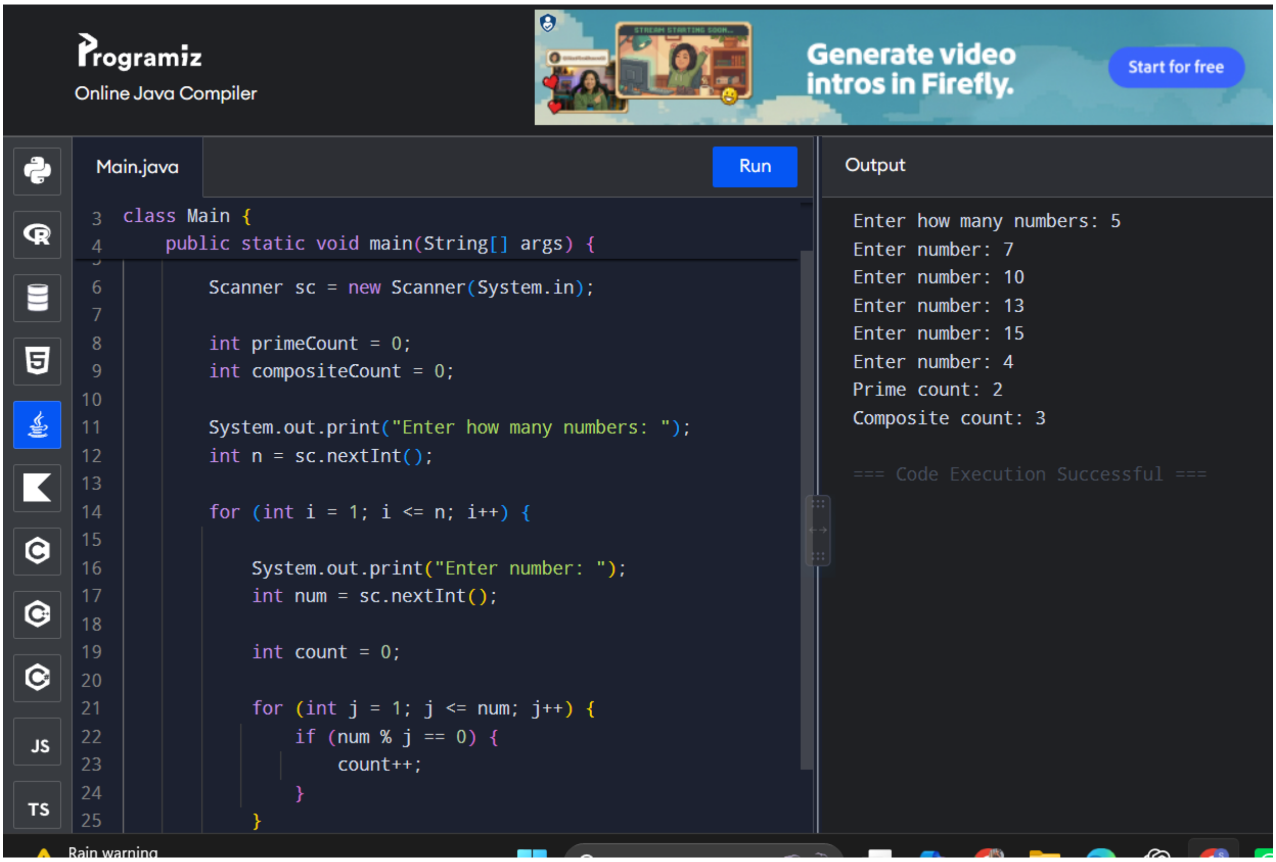
}



2 Find the count of prime numbers and composite numbers among the given inputs

```
 import java.util.Scanner;


class Main {
    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);


        int primeCount = 0;
        int compositeCount = 0;


        System.out.print("Enter how many numbers: ");
        int n = sc.nextInt();


        for (int i = 1; i <= n; i++) {


            System.out.print("Enter number: ");
```

```
            int num = sc.nextInt();

            int count = 0;

            for (int j = 1; j <= num; j++) {
                if (num % j == 0) {
                    count++;
                }
            }

            if (count == 2) {
                primeCount++;
            } else if (count > 2) {
                compositeCount++;
            }
        }

        System.out.println("Prime count: " + primeCount);
        System.out.println("Composite count: " + compositeCount);
    }
}
```



3 In an organization they decide to give bonus to all the employees on New Year. A 5% bonus on salary is given to the grade A workers and 10% bonus on salary to the grade B workers. Write a program to enter the salary and grade of the employee. If the salary of the employee is less than $10,000 then the employee gets an extra 2% bonus on salary Calculate the bonus that has to be given to the employee and print the salary that the employee will get. Sample Input & Output: Enter the grade of the employee: B Enter the employee salary: 50000 Salary=50000 Bonus=5000.0 Total to be paid:55000.0 give java code simple java code

```
import java.util.Scanner;


class Main {

  public static void main(String[] args) {


    Scanner sc = new Scanner(System.in);


    System.out.print("Enter the grade of the employee: ");
```

```java
        char grade = sc.next().charAt(0);

        System.out.print("Enter the employee salary: ");
        double salary = sc.nextDouble();

        double bonus = 0;

        if (grade == 'A') {
            bonus = salary * 0.05;
        } else if (grade == 'B') {
            bonus = salary * 0.10;
        }

        if (salary < 10000) {
            bonus = bonus + salary * 0.02;
        }

        double total = salary + bonus;

        System.out.println("Salary=" + salary);
        System.out.println("Bonus=" + bonus);
        System.out.println("Total to be paid:" + total);
    }
}
```

Main.java

Run

```java
import java.util.Scanner;

class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the grade of the employee: ");
        char grade = sc.next().charAt(0);

        System.out.print("Enter the employee salary: ");
        double salary = sc.nextDouble();

        double bonus = 0;

        if (grade == 'A') {
            bonus = salary * 0.05;
        } else if (grade == 'B') {
            bonus = salary * 0.10;
        }

        if (salary < 10000) {
            bonus = bonus + salary * 0.02;
```

Output

```
Enter the grade of the employee: b
Enter the employee salary: 50000
Salary=50000.0
Bonus=0.0
Total to be paid:50000.0

=== Code Execution Successful ===
```